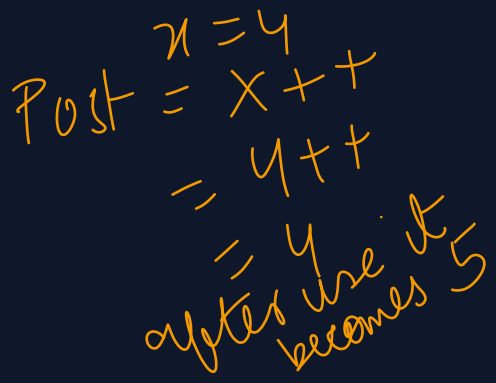
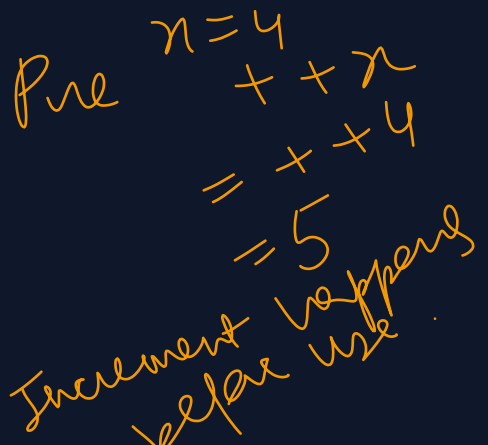
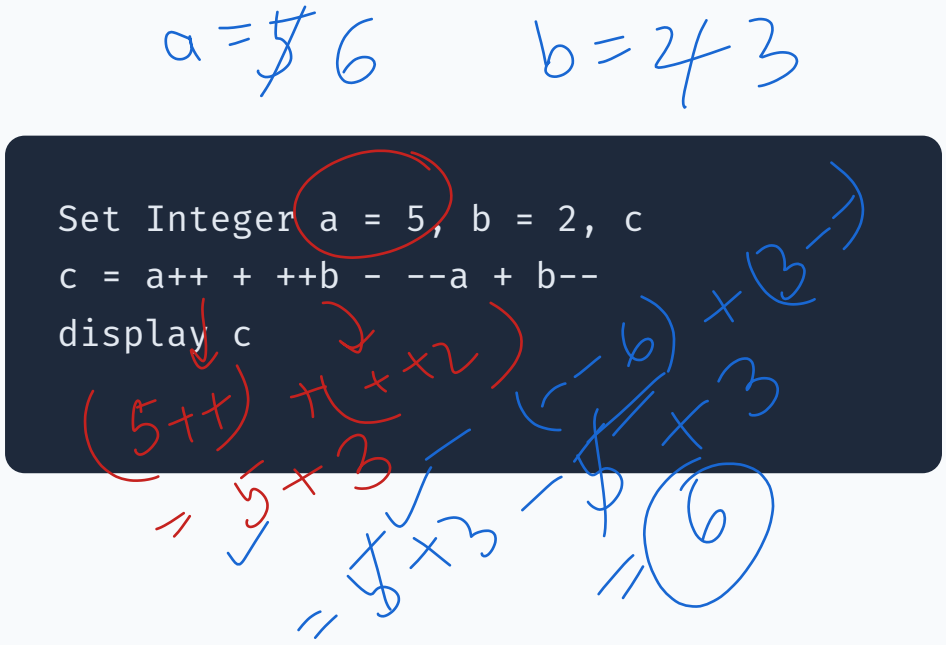
Infosys Pseudo Code
Pre and Post Increment/Decrement



BASED ON LATEST OCTOBER PATTERN



QUESTION 01



```
Set Integer a = 5, b = 2, c
c = a++ + ++b - --a + b--
display c
```

What will be the output?

- 5
- 6
- 7
- 8

SOLUTION 01

```
Set Integer a = 5, b = 2, c
c = a++ + ++b - --a + b--
display c
```

6

EXECUTION DRY RUN

Step 1: Post-increment a returns 5.

Step 2: Pre-increment b returns 3.

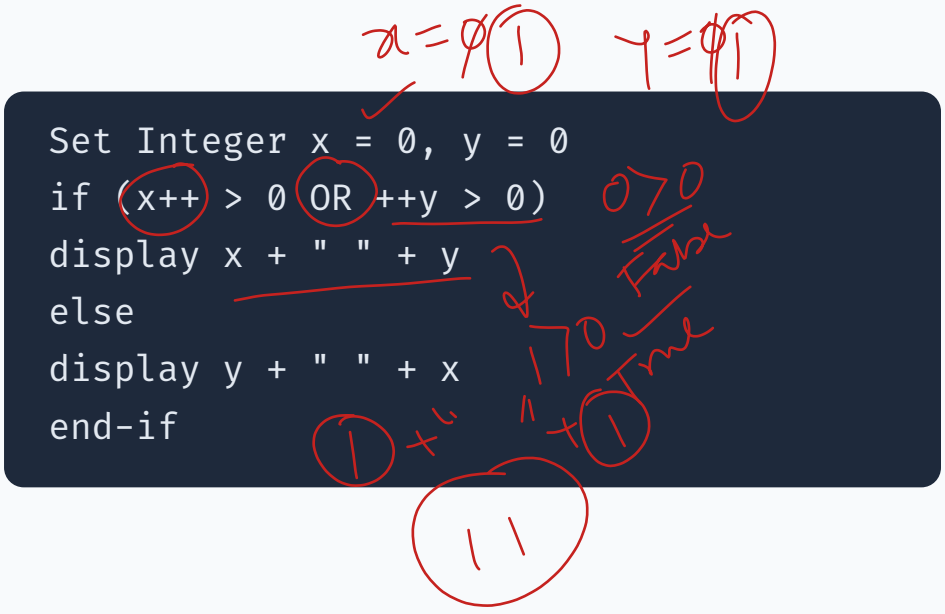
Step 3: Pre-decrement a returns 5.

Step 4: Post-decrement b returns 3.

Calculate: Formula yields 5 + 3 - 5 + 3 = 6.



QUESTION 02



What will be the output?

- 0 0
- 1 0
- 0 1
- 1 1

SOLUTION 02

```
Set Integer x = 0, y = 0
if (x++ > 0 OR ++y > 0)
display x + " " + y
else
display y + " " + x
end-if
```

1 1

EXECUTION DRY RUN

Left OR: x++ returns 0. Condition fails.

Update X: Post-increment sets x to 1.

Right OR: ++y sets y to 1. Condition passes.

Result: True overall condition outputs 1 1.

QUESTION 03

```
Set Integer p = 5, q = 10
if (++p < 5 AND q-- > 9)
display p
else
display p + " " + q
end-if
```

What will be the output?

- 6 10
- 6 9
- 5 10
- 5 9

SOLUTION 03

```
Set Integer p = 5, q = 10
if (++p < 5 AND q-- > 9)
display p
else
display p + " " + q
end-if
```

6 10

EXECUTION DRY RUN

Left AND: ++p sets p to 6. Evaluation fails.

Bypass: Right operand evaluation short-circuits.

State: Variable q remains exactly 10.

Result: Else branch outputs 6 10.

QUESTION 04

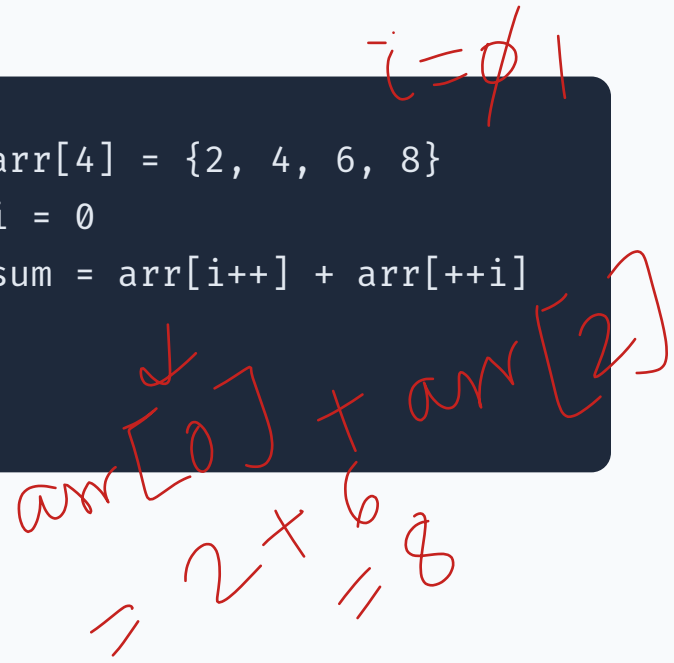
```
Set Integer arr[4] = {2, 4, 6, 8}
Set Integer i = 0
Set Integer sum = arr[i++] + arr[++i]
display sum
```

What will be the value of sum?

- 6
- 8
- 10
- 14

SOLUTION 04

```
Set Integer arr[4] = {2, 4, 6, 8}
Set Integer i = 0
Set Integer sum = arr[i++] + arr[++i]
display sum
```



8

EXECUTION DRY RUN

Term 1: Fetch index 0 outputs value 2.

Update: Pointer i post-increments to 1.

Term 2: Pre-increment sets i to 2. Fetch value 6.

Calculate: Sum totals 2 + 6 yielding 8.

QUESTION 05

```
Set Integer val = 13
Set Integer res = (val++ MOD 5) +
(++val MOD 3)
display res
```

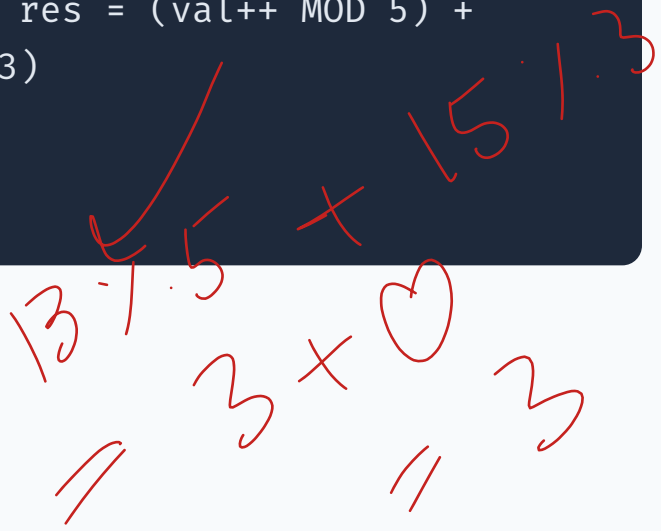
What will be the output?

- 2
- 3
- 4
- 5

SOLUTION 05



```
Set Integer val = 13
Set Integer res = (val++ MOD 5) +
(++val MOD 3)
display res
```



3

EXECUTION DRY RUN

Term 1: Modulo 13 by 5 yields 3.

Update: Variable val post-increments to 14.

Term 2: Pre-increment sets val to 15. Modulo yields 0.

Calculate: Addition 3 + 0 yields exactly 3.

QUESTION 06

```
Set Integer count = 5
while (count-- > 2)
do
display --count
end-while
```

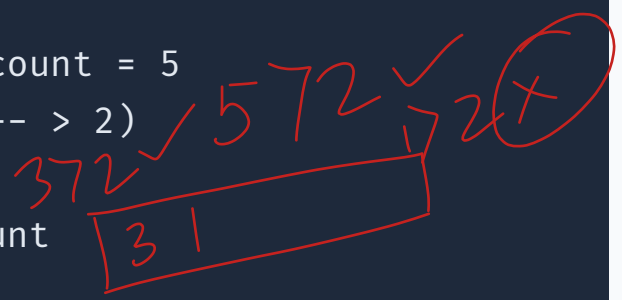
What will be the output?

4 2 0	3 1
3 2	4 3 2

SOLUTION 06



```
Set Integer count = 5
while (count-- > 2)
do
display --count
end-while
```



```
3 1
```

EXECUTION DRY RUN

Iter 1: Condition passes. Count drops to 4. Prints 3.

Iter 2: Condition passes. Count drops to 2. Prints 1.

Iter 3: Condition fails 1 > 2. Loop terminates.

Result: Sequential outputs print 3 1.

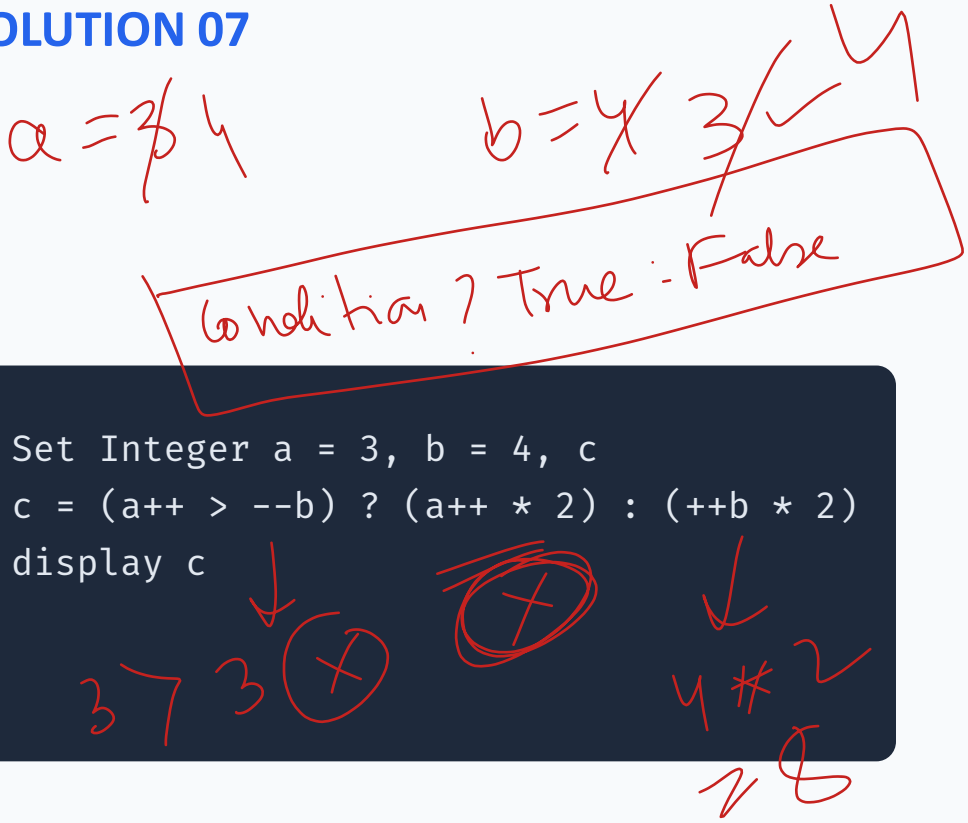
QUESTION 07

```
Set Integer a = 3, b = 4, c
c = (a++ > --b) ? (a++ * 2) : (++b * 2)
display c
```

What will be the value of c?

- 6
- 8
- 10
- 12

SOLUTION 07




```
Set Integer a = 3, b = 4, c
c = (a++ > --b) ? (a++ * 2) : (++b * 2)
display c
```


8

EXECUTION DRY RUN

Condition: Compare 3 > 3 yields False.

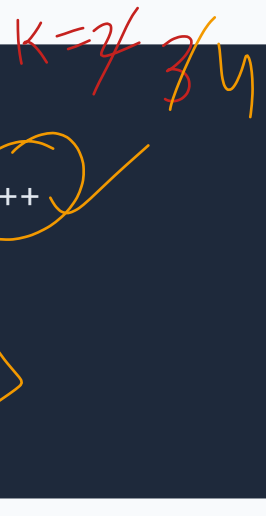
Update: Variable a post-increments to 4.

Branch: False logic executes right side.

Calculate: Variable b increments 4. Result 4 * 2 = 8.

QUESTION 08

```
Set Integer k = 2
switch(k++)
case 2: display "Two " k++
        break
case 3: display "Three "
        break
end-switch
display k
```



What will be the output?

- Two 3
- Two 4
- Three 3
- Three 4

SOLUTION 08

```
... switch(k++) case 2: display "Two "
k++ ...
```

Two 4

EXECUTION DRY RUN

Switch: Fetch 2 navigates Case 2 directly.

Update: Post-increment sets variable k to 3.

Case 2: Prints string Two. Increments k to 4.

Result: Final instruction displays value 4.

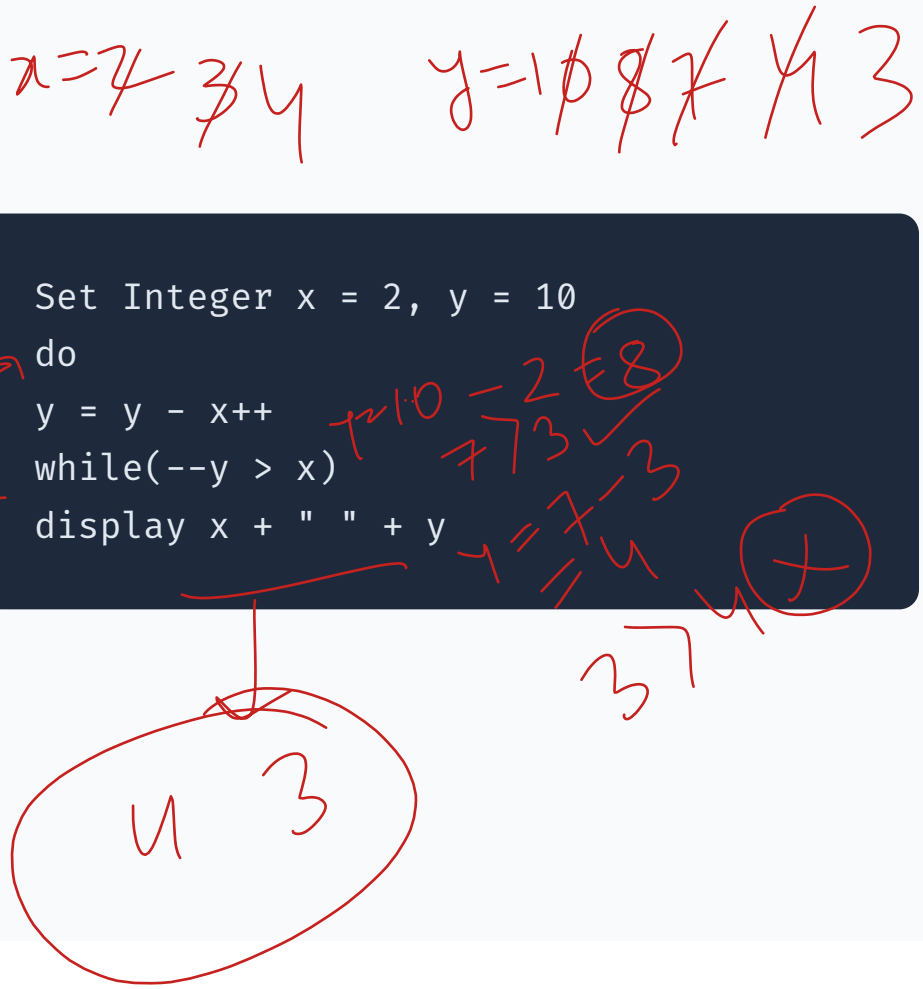
QUESTION 09

```
Set Integer x = 2, y = 10
do
y = y - x++
while(--y > x)
display x + " " + y
```

What will be the output?

- 38
- 43
- 44
- 37

SOLUTION 09



```
Set Integer x = 2, y = 10
do
y = y - x++
while(--y > x)
display x + " " + y
```

4 3

EXECUTION DRY RUN

Iter 1: Calculate y = 8. Update x = 3. Check 7 > 3 passes.

Iter 2: Calculate y = 4. Update x = 4. Check 3 > 4 fails.

Result: Loop ends tracking values x=4 y=3.

QUESTION 10

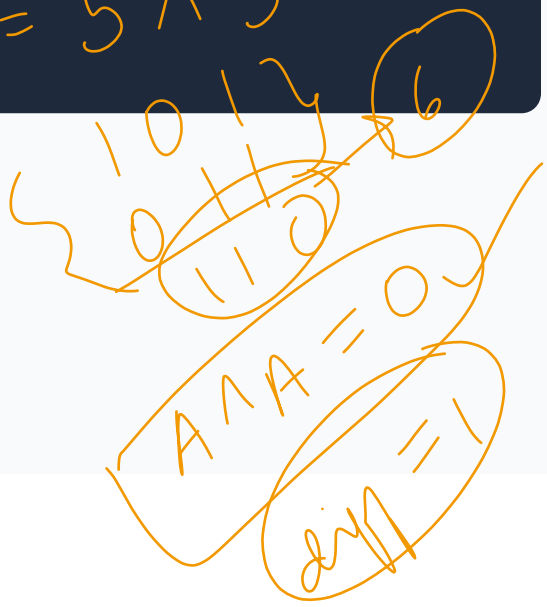
```
Set Integer a = 5, b = 2
Set Integer c = a++ ^ ++b
display c
```

What will be the value of c?

4	5
6	7

SOLUTION 10

```
Set Integer a = 5, b = 2
Set Integer c = a++ ^ ++b
display c
```



6

EXECUTION DRY RUN

Term 1: Fetch variable a yields 5.

Term 2: Pre-increment sets variable b yielding 3.

Bitwise: XOR operation compares binary 101 and 011.

Result: Output measures decimal 6 reliably.

QUESTION 11

```
Set Integer m = 1
Set Integer result = m++ * 2 + ++m * 3
- m-- * 4
display result
```

What will be the value of result?

-1	0
1	2

SOLUTION 11

```
Set Integer m = 1
Set Integer result = m++ * 2 + ++m * 3
- m-- * 4
display result
```

-1

EXECUTION DRY RUN

Term 1: Use 1 yields 2. Update m = 2.

Term 2: Set 3 yields 9. Update m = 3.

Term 3: Use 3 yields 12. Update m = 2.

Calculate: Combine 2 + 9 - 12 yielding -1.

QUESTION 12

```
Set Integer i = 1, sum = 0
while(i++ < 4)
do
sum = sum + ++i
end-while
display sum
```

What will be the output?

- 5
- 8
- 12
- 15

SOLUTION 12

```
Set Integer i = 1, sum = 0
while(i++ < 4)
do
sum = sum + ++i
end-while
display sum
```

8

EXECUTION DRY RUN

Iter 1: Check 1 < 4. Inside i=3 sum=3.

Iter 2: Check 3 < 4. Inside i=5 sum=8.

Iter 3: Check 5 < 4 fails instantly.

Result: Loop exits displaying sum 8.

QUESTION 13

```
Set Integer x = 14
Set Integer y = x++ / 3 + ++x / 4
display y
```

What will be the value of y?

7	8
9	10

SOLUTION 13

```
Set Integer x = 14
Set Integer y = x++ / 3 + ++x / 4
display y
```

8

EXECUTION DRY RUN

Term 1: Execute 14 / 3 yielding 4.

Update: Variable x increments 15 internally.

Term 2: Increment x to 16. Execute 16 / 4 yielding 4.

Calculate: Evaluate 4 + 4 targeting 8.

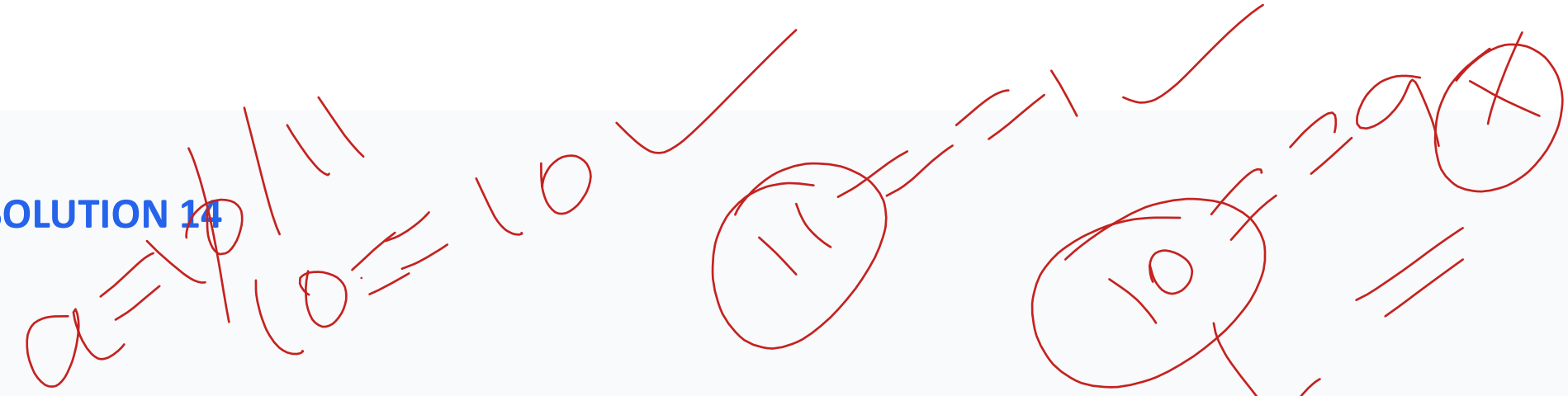
QUESTION 14

```
Set Integer a = 10, b = 10, c = 10
if (a++ == 10 AND ++b == 11 AND c-- ==
9)
display "If"
else
display "Else " + a + " " + b + " " + c
end-if
```

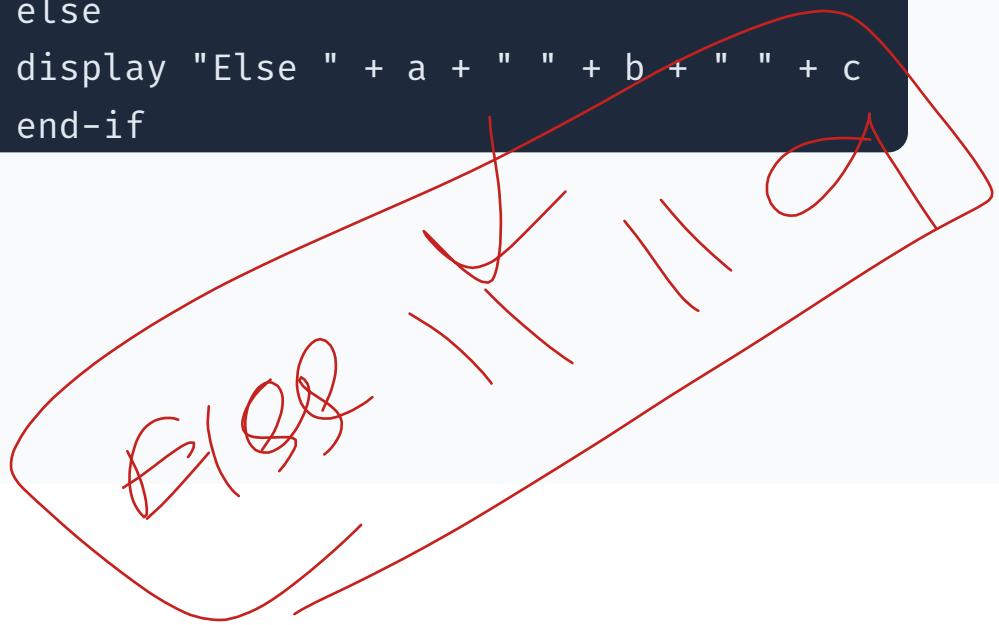
What will be the output?

- If
- Else 10 11 9
- Else 11 11 9
- Else 11 11 10

SOLUTION 14



```
Set Integer a = 10, b = 10, c = 10
if (a++ == 10 AND ++b == 11 AND c-- ==
9)
display "If"
else
display "Else " + a + " " + b + " " + c
end-if
```



Else 11 11 9

EXECUTION DRY RUN

Cond 1: Check 10 == 10 passes. Variable a becomes 11.

Cond 2: Check 11 == 11 passes. Variable b becomes 11.

Cond 3: Check 10 == 9 fails. Variable c becomes 9.

Result: Logic triggers Else printing updated states.

QUESTION 15

```
Set Integer i = 0, j = 0, count = 0
while(i++ < 2) do
while(++j < 3) do
count++
end-while
end-while
display count
```

What will be the output?

- 2
- 3
- 4
- 6

SOLUTION 15

```
Set Integer i = 0, j = 0, count = 0
while(i++ < 2) do
while(++j < 3) do
count++
end-while
end-while
display count
```

2

EXECUTION DRY RUN

Outer 1: Uses i=0 passes. Inner loop executes.

Inner: Variables j increments 1 then 2. Count reaches 2.

Outer 2: Uses i=1 passes. Inner loop skips totally.

Result: Outer fails i=2. Count remains exactly 2.

End of Practice Session

MASTERING OPERATOR PRECEDENCE